# STACKS, MODULI OF CURVES: LEMMA-GENUS

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 1127-1139 of sources/stacks-moduli-curves/pinned-original-source.tex; proof span is lines 1141-1165, the complete author proof (23 body lines).

*Editorial note.* Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 21 of queue_moduli-curves.json (pb=974, stmt 1127-1139, proof 1141-1165).

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma 02UZ: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 1165–1165. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0E6I: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 1078–1088. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0BCD: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 1165–1165. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0CTM: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 1165–1165. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0E6B: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 1165–1165. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $f : X \to S$ be a family of curves such that $\kappa(s) = H^0(X_s, \mathcal{O}_{X_s})$ for all $s \in S$, i.e., the classifying morphism $S \to \mathcal{C}urves$ factors through $\mathcal{C}urves^{h0,1}$ (Lemma 0E6I). Then*

*Date*: stacks-project:moduli-curves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) $f_*\mathcal{O}_X = \mathcal{O}_S$ *and this holds universally,*
(2) $R^1 f_*\mathcal{O}_X$ *is a finite locally free* $\mathcal{O}_S$*-module,*
(3) *for any morphism* $h : S' \to S$ *if* $f' : X' \to S'$ *is the base change, then* $h^*(R^1 f_*\mathcal{O}_X) = R^1 f'_*\mathcal{O}_{X'}$.

*Proof.* We apply Derived Categories of Spaces, Lemma 0E6B. This proves part (1). It also implies that locally on $S$ we can write $Rf_*\mathcal{O}_X = \mathcal{O}_S \oplus P$ where $P$ is perfect of tor amplitude in $[1, \infty)$. Recall that formation of $Rf_*\mathcal{O}_X$ commutes with arbitrary base change (Derived Categories of Spaces, Lemma 0CTM). Thus for $s \in S$ we have

$$H^i(P \otimes^{\mathbf{L}}_{\mathcal{O}_S} \kappa(s)) = H^i(X_s, \mathcal{O}_{X_s}) \text{ for } i \geq 1$$

This is zero unless $i = 1$ since $X_s$ is a 1-dimensional Noetherian scheme, see Cohomology, Proposition 02UZ. Then $P = H^1(P)[-1]$ and $H^1(P)$ is finite locally free for example by More on Algebra, Lemma 0BCD. Since everything is compatible with base change we also see that (3) holds. □